

BioHub Cell Tracking Report

Sample: memory_profile

Frames: 6

--- Dashboard ---

Detected cells: 14

Tracks: 4

Divisions: 1

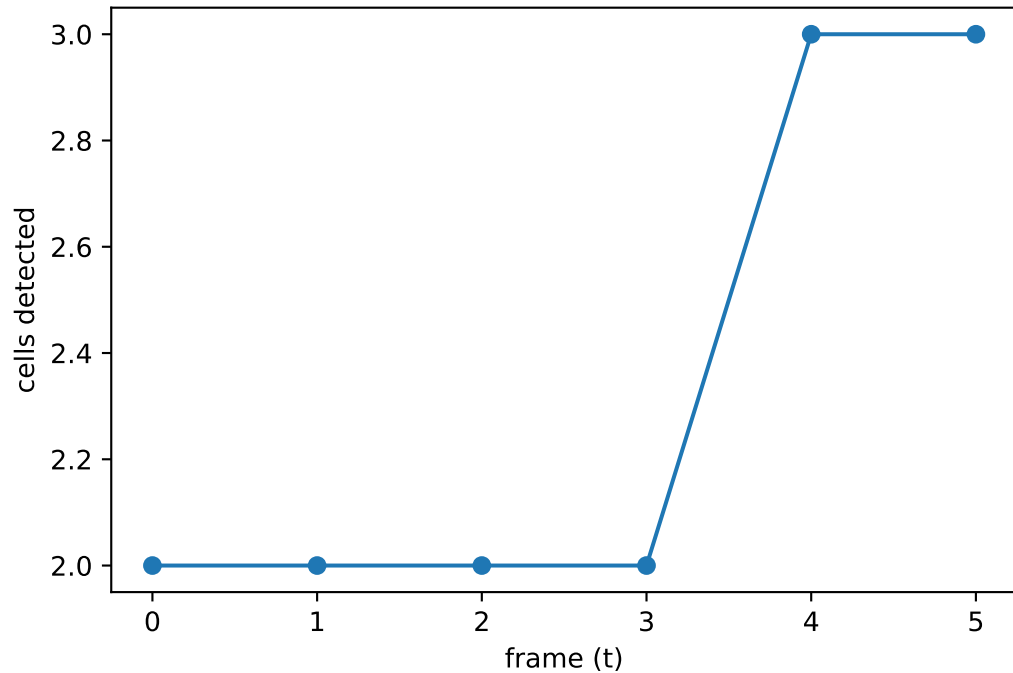
Avg. lifetime: 3.5 frames

Avg. velocity: 3.747 px/frame

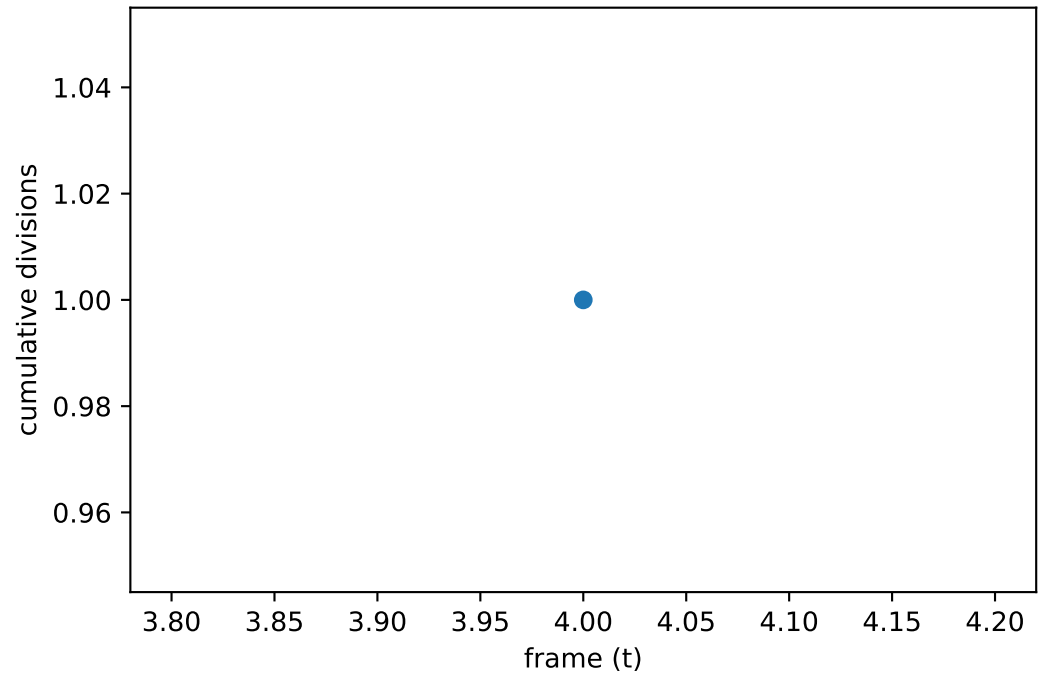
Method

Cells were detected per frame via Gaussian smoothing ($\sigma=2$) followed by local maximum detection (threshold=800, min. distance=12px) on each frame's maximum-intensity projection. Candidate detections were classified by a trained CNN (CellCNN). Frame-to-frame linking used optimal (Hungarian) bipartite assignment (max. distance=25px); cell divisions were detected via a second assignment pass on unmatched detections (max. distance=37.5px), each daughter assigned a new track ID linked to its parent.

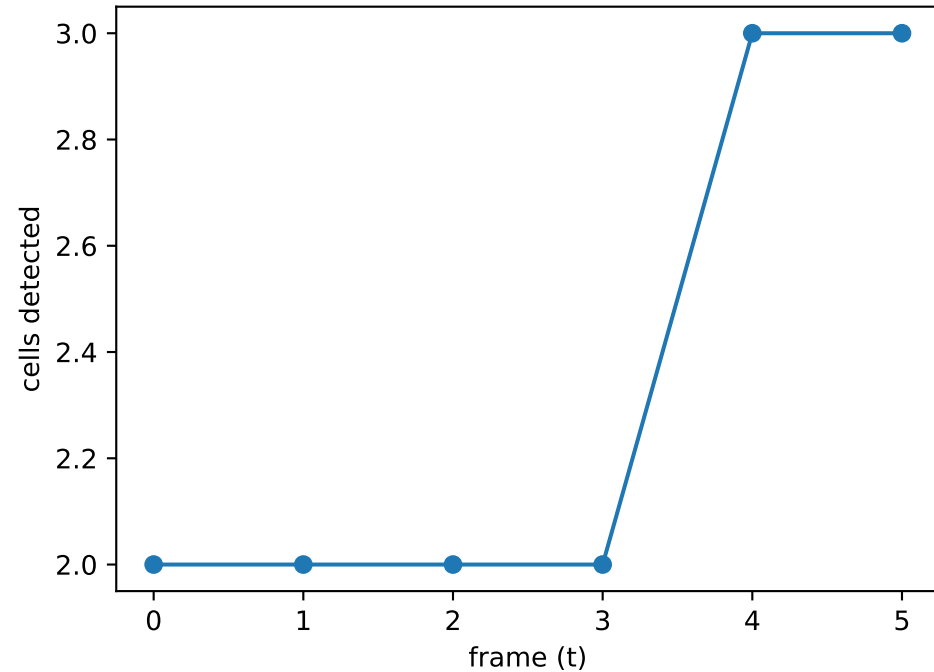
Cells per frame



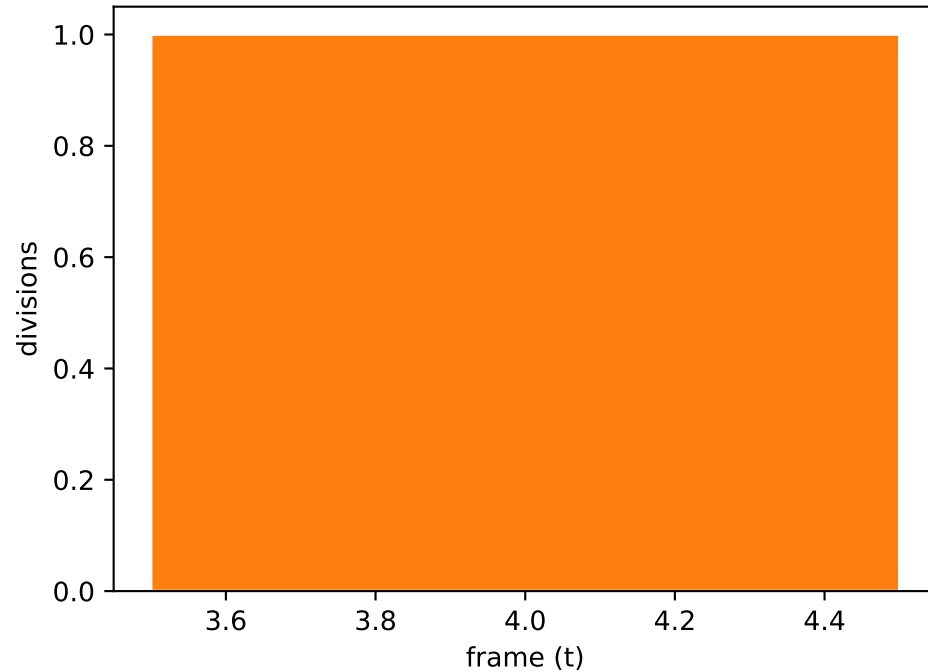
Divisions over time (total: 1)



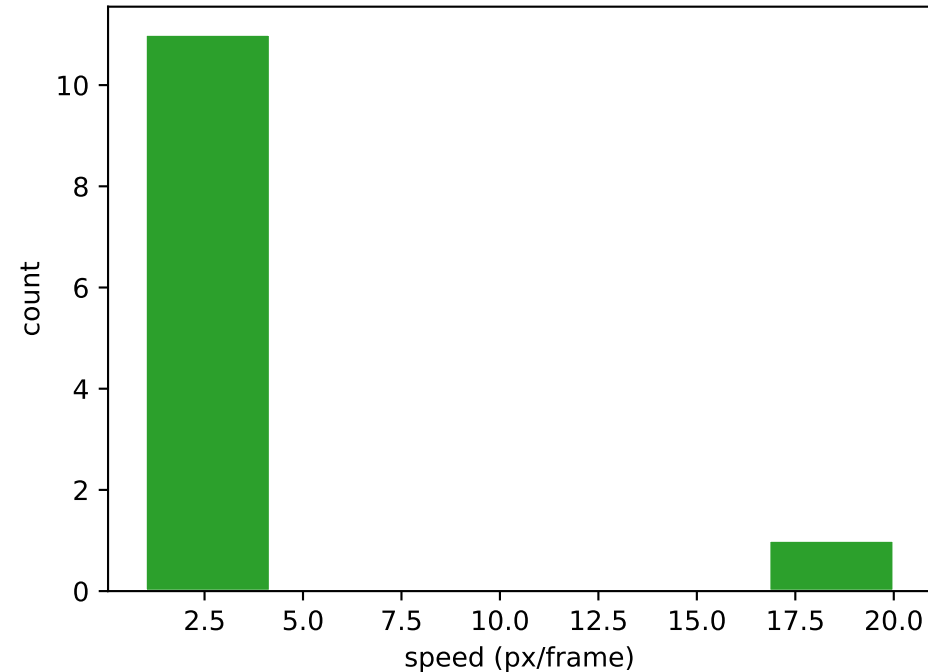
Cell count over time



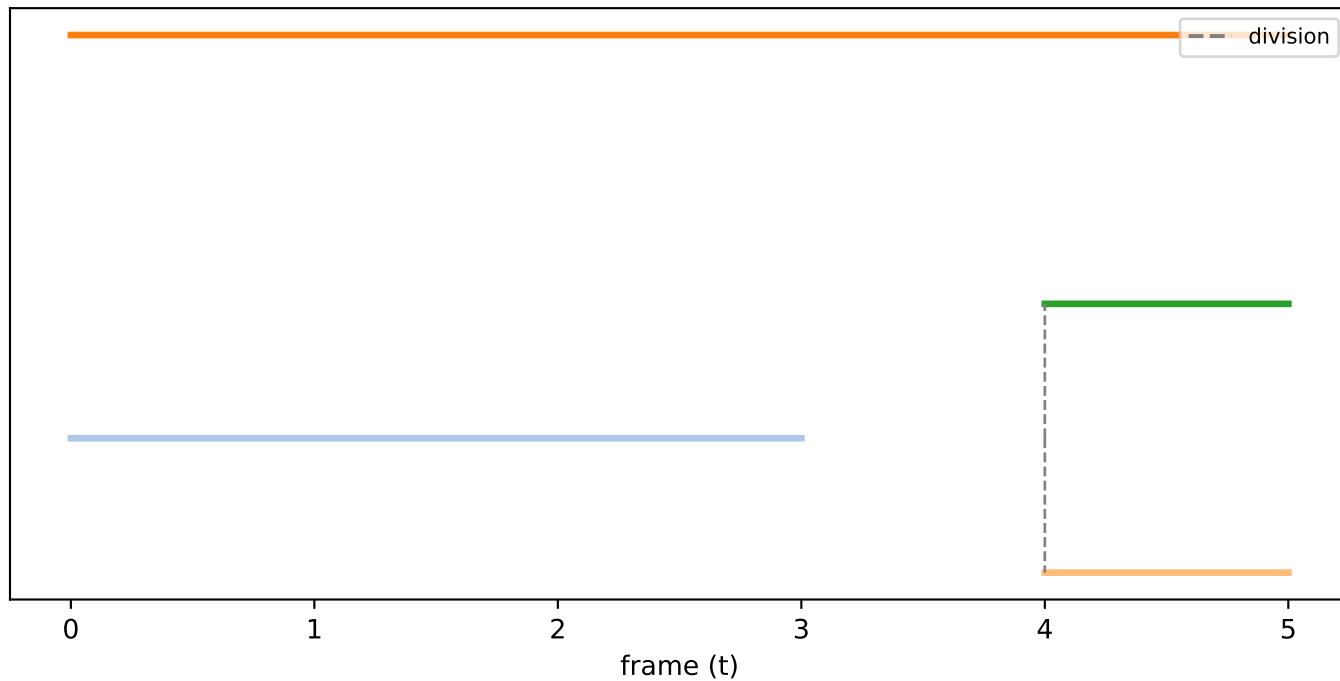
Divisions over time (total: 1)



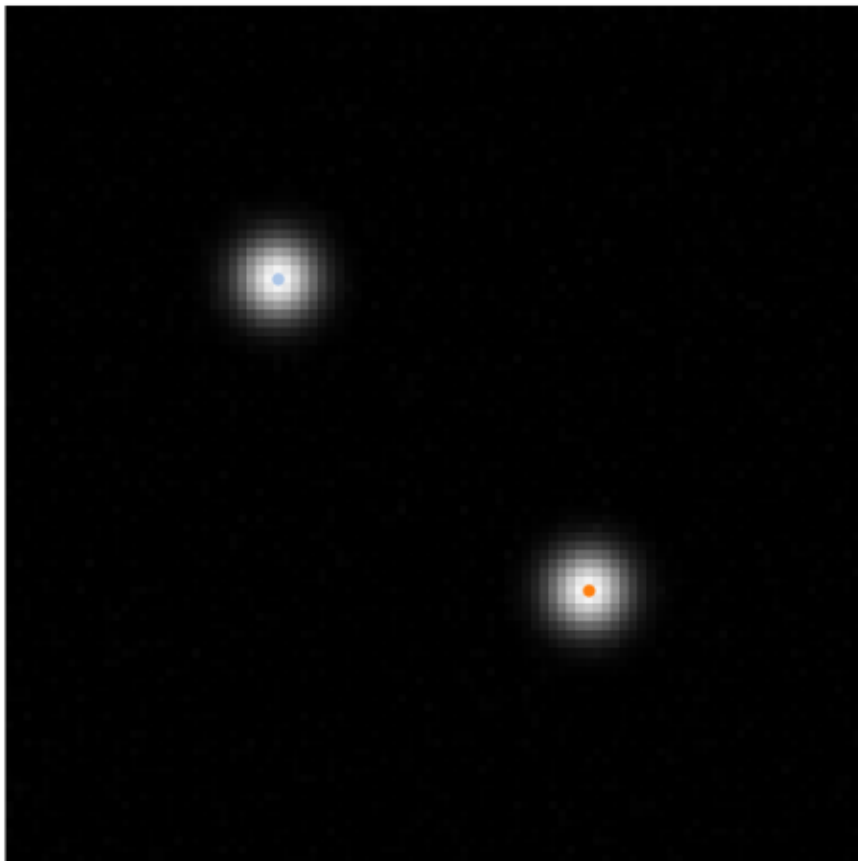
Velocity distribution



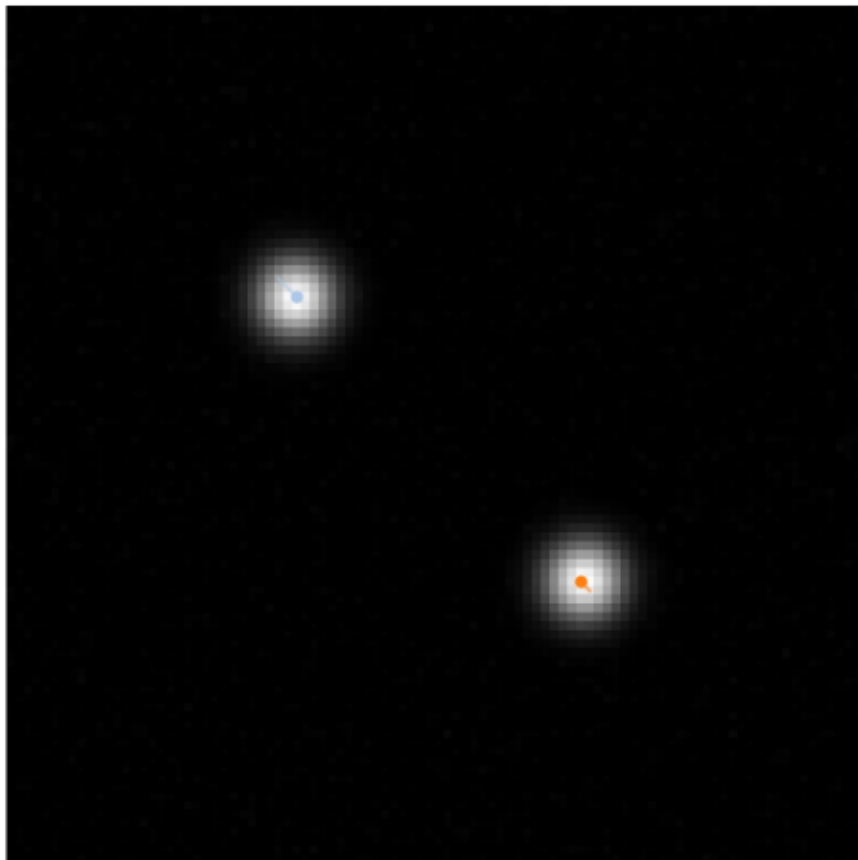
Lineage tree — 4 tracks, 1 division(s)



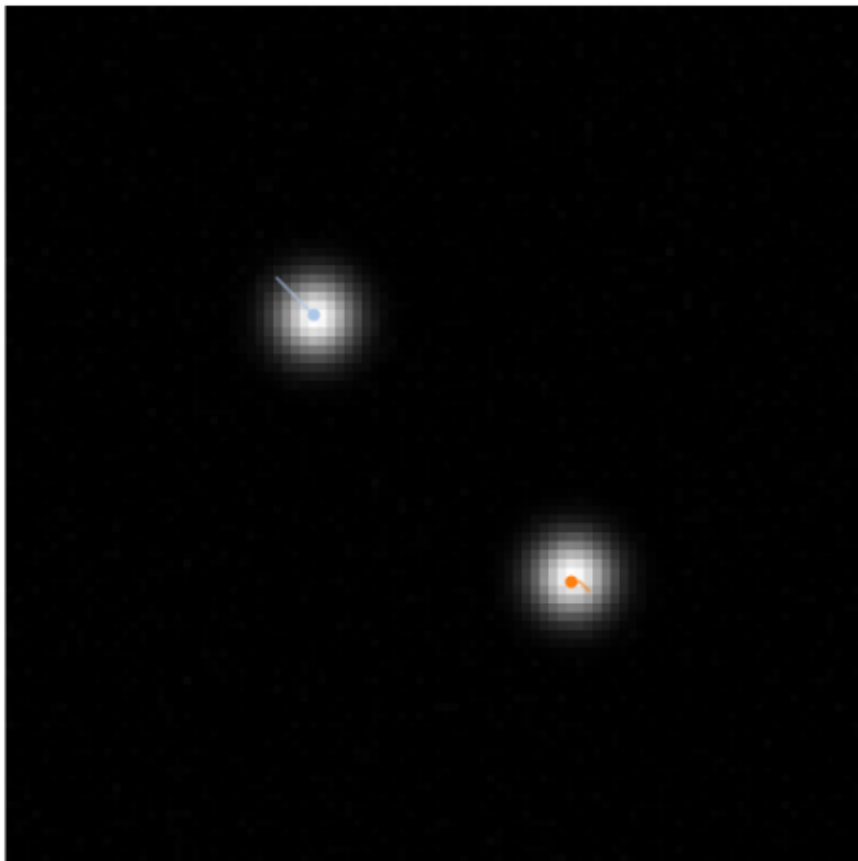
$t = 0$ (2 tracks visible, 0 division(s) this frame)



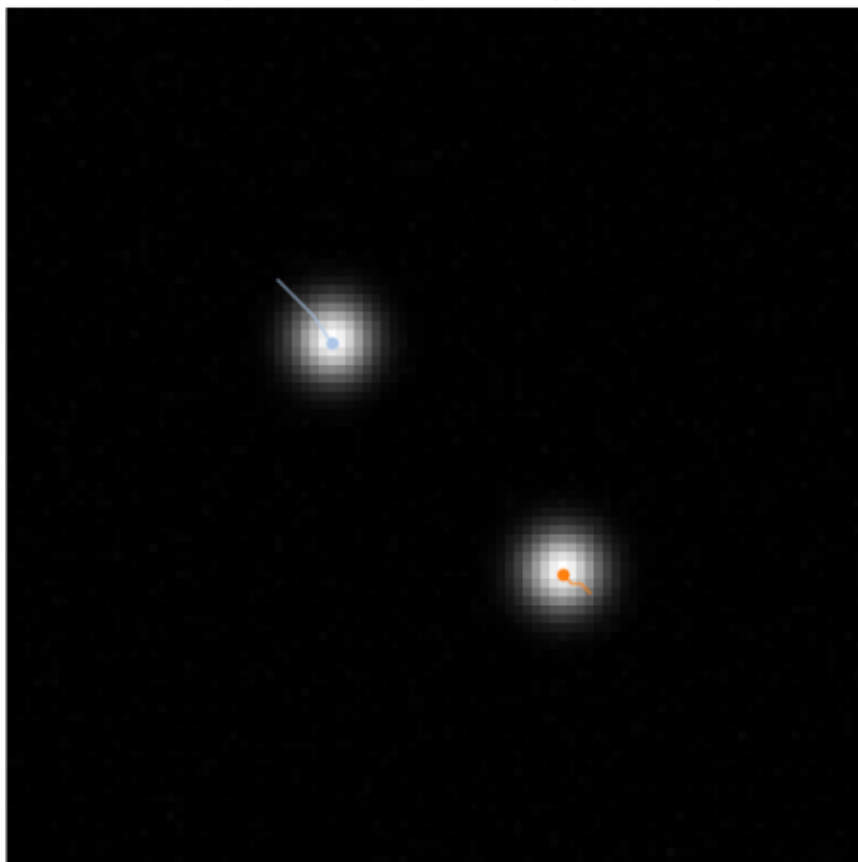
t = 1 (2 tracks visible, 0 division(s) this frame)



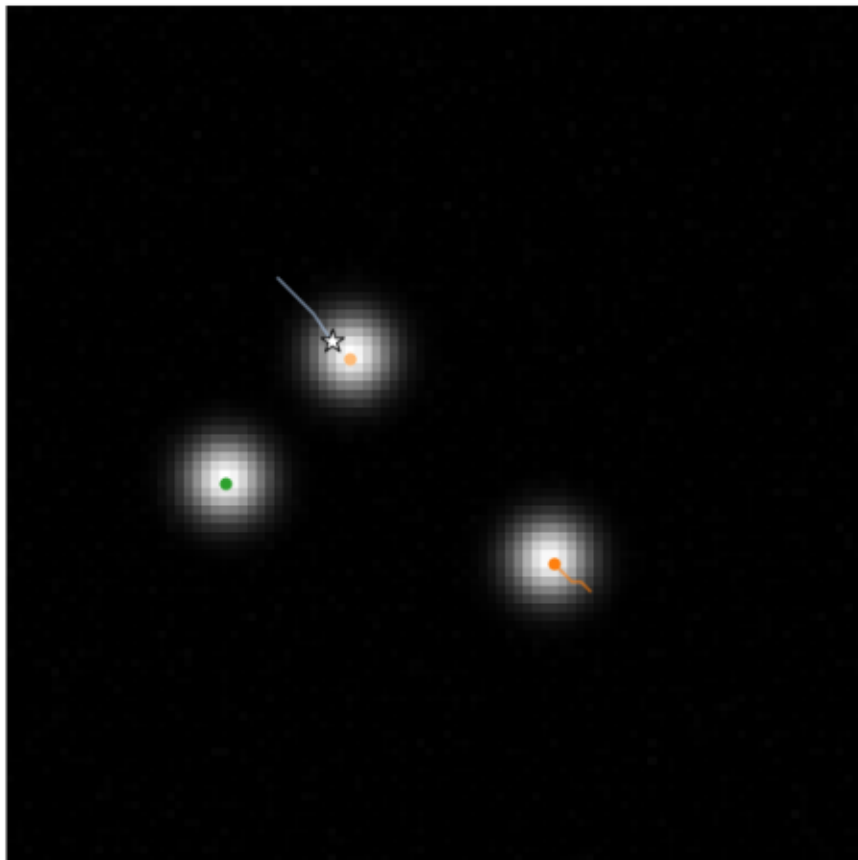
t = 2 (2 tracks visible, 0 division(s) this frame)



t = 3 (2 tracks visible, 0 division(s) this frame)



$t = 4$ (4 tracks visible, 1 division(s) this frame)



$t = 5$ (4 tracks visible, 0 division(s) this frame)

